

NARAN BULAG, China

WMO: 530830

Lat: 44.6175N

Lon: 114.1523E

Elev: 3881

StdP: 12.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-19.5	-29.8	1.2	-21.2	-26.4	1.5	-17.5	22.9	4.9	20.3	3.1	2.9	320	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	89.5	60.9	85.9	59.9	82.7	58.8	66.0	77.6	64.0	76.7	62.6	75.2	10.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	97.7	70.1	60.3	90.3	68.7	58.2	83.8	67.5	33.1	77.1	31.4	76.3	30.2	75.2	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.6	22.2	19.2	DB	-27.3	94.8	4.1	3.5	-30.2	97.3	-32.6	99.4	-34.9	101.4	-37.9	103.9	
			WB	-27.5	68.7	4.1	2.7	-30.4	70.6	-32.8	72.2	-35.1	73.7	-38.1	75.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	-4.3	4.7	22.2	40.2	53.5	64.7	70.3	66.6	55.0	37.6	16.7	0.2
	DBStd	27.40	8.66	10.39	11.02	9.58	8.51	6.66	5.42	5.98	8.04	9.32	11.17	8.47
	HDD50	7176	1685	1268	862	315	63	0	0	0	42	396	1000	1545
	HDD65	11041	2150	1688	1326	743	367	86	12	49	309	851	1450	2010
	CDD50	1982	0	0	0	22	170	442	629	516	193	10	0	0
	CDD65	372	0	0	0	0	9	77	176	100	10	0	0	0
	CDH74	4063	0	0	0	11	232	891	1763	1018	147	1	0	0
	CDH80	1326	0	0	0	0	57	274	651	320	24	0	0	0
(14) Wind	WSAvg	8.1	5.6	6.6	9.1	10.8	11.2	9.0	8.1	7.7	8.1	8.3	6.9	5.8
(15) Precipitation	PrecAvg	8.7	0.0	0.1	0.1	0.2	0.6	1.4	2.5	2.5	0.8	0.3	0.1	0.1
	PrecMax	14.7	0.2	0.2	0.6	1.5	2.0	3.7	5.2	5.6	2.0	0.8	0.4	0.2
	PrecMin	4.9	0.0	0.0	0.0	0.0	0.1	0.2	0.7	0.2	0.0	0.0	0.0	0.0
	PrecStd	2.4	0.0	0.0	0.1	0.3	0.5	0.9	1.1	1.3	0.6	0.2	0.1	0.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	37.5	57.9	75.1	86.7	90.9	95.0	93.0	83.2	68.7	48.4	28.2
		MCWB	18.6	28.2	39.3	47.7	54.9	59.2	63.4	62.7	56.9	48.6	36.1	23.5
	2%	DB	17.4	30.8	50.0	67.9	80.3	86.7	90.6	87.2	78.4	62.9	42.3	20.5
		MCWB	14.1	24.6	35.6	45.2	52.0	58.1	62.4	60.6	55.1	46.0	32.9	17.2
	5%	DB	12.8	26.0	44.6	62.8	75.4	83.0	86.8	83.2	73.8	57.5	37.0	16.2
		MCWB	10.4	21.5	32.1	42.6	50.5	57.1	61.5	59.6	52.7	42.4	29.4	13.6
	10%	DB	8.7	20.8	39.3	57.8	71.0	79.2	83.4	79.5	69.8	52.7	32.4	12.4
		MCWB	6.9	17.0	29.3	40.3	48.6	56.4	60.9	58.8	51.5	40.0	26.7	10.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	29.3	40.1	49.5	57.4	63.5	69.5	67.7	61.8	51.1	37.3	23.1
		MCDB	23.3	35.9	55.5	71.2	78.0	77.5	80.9	81.2	75.8	63.7	47.7	28.1
	2%	WB	14.2	25.1	35.8	46.0	54.5	61.6	67.0	65.1	57.8	46.8	32.8	17.3
		MCDB	17.4	30.6	48.9	65.5	72.9	76.1	78.5	76.7	69.8	60.2	41.4	20.4
	5%	WB	10.5	21.1	32.6	43.5	52.3	60.1	65.2	63.3	55.6	43.6	29.9	13.7
		MCDB	12.8	25.9	43.3	61.2	70.1	74.8	77.7	74.6	68.9	55.8	36.3	16.1
	10%	WB	7.0	17.1	29.7	40.8	50.1	58.7	63.6	61.6	53.6	40.6	27.0	10.4
		MCDB	8.6	20.6	38.8	57.0	67.8	73.7	76.8	73.7	66.4	51.5	32.5	12.4
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	20.4	22.1	23.5	24.3	21.6	20.7	21.4	22.5	21.8	19.7	17.2
		MCDBR	21.4	24.5	27.7	30.8	32.2	27.5	26.3	26.6	27.4	28.1	24.2	20.6
		MCWBR	18.8	19.1	17.1	15.8	14.7	9.6	8.1	9.0	11.8	15.2	16.6	17.7
	5% WB	MCDBR	21.2	24.0	26.9	28.6	27.0	21.4	19.6	20.6	22.7	26.0	23.2	20.1
		MCWBR	18.7	19.0	17.2	15.3	13.3	8.0	8.2	8.8	11.0	15.2	16.3	17.4
(40) Clear-Sky Solar Irradiance	taub		0.256	0.287	0.328	0.379	0.437	0.459	0.420	0.404	0.359	0.320	0.279	0.259
	taud		2.323	2.245	2.178	2.085	1.964	2.014	2.198	2.205	2.266	2.304	2.312	2.292
	Ebn at Noon		277	285	286	279	265	259	268	268	272	268	263	261
	Edh at Noon		25	32	40	48	56	54	44	43	37	30	25	23
(44) All-Sky Solar Radiation	RadAvg		667	1026	1518	1915	2122	2069	2033	1885	1562	1133	744	549
	RadStd		71	76	92	98	126	135	76	68	80	66	56	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	-2.77	N/A	N/A	-1.17	N/A	N/A	N/A	N/A

Nomenclature: See separate page